

Pandemic-Driven Technology During COVID-19: Experiences of Older Adults

Raghad A. Alharbi
University of Jeddah
Jeddah, Saudi Arabia

Farah S. Alamri
University of Jeddah
Jeddah, Saudi Arabia

Faisal T. Altayyari
University of Jeddah
Jeddah, Saudi Arabia

Sultan A. Alharthi
University of Jeddah
Jeddah, Saudi Arabia

ABSTRACT

Pandemic-driven technology and mobile health (mHealth) applications become more apparent with the COVID-19 pandemic. Older adults are facing major issues using these applications due to their lack of understanding of the technology and hesitation to adopt them, reducing the benefits of such applications. To further explore these issues, we conducted an online survey with older adults' families ($n = 397$), preformed online interviews with older adults ($n = 5$), and undertook observations focused on their experiences. We found that older users of such social pandemic tracking technologies suffered with using these applications by themselves, which resulted in an increased dependence on others to help them with their basic daily social activities, increasing their chances of physically interacting and contracting the coronavirus. Such results make us question the level of technology literacy we reached within our communities, especially with the increased integration of digital technology into all areas of our lives.

CCS CONCEPTS

• **Human-centered computing** → **Ethnographic studies**; *Empirical studies in HCI*.

KEYWORDS

Pandemic-driven technology, older adults, COVID-19, mHealth, social interaction.

ACM Reference Format:

Raghad A. Alharbi, Faisal T. Altayyari, Farah S. Alamri, and Sultan A. Alharthi. 2021. Pandemic-Driven Technology During COVID-19: Experiences of Older Adults. In *Companion Publication of the 2021 Conference on Computer Supported Cooperative Work and Social Computing (CSCW '21 Companion)*, October 23–27, 2021, Virtual Event, USA. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3462204.3481769>

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.
CSCW '21 Companion, October 23–27, 2021, Virtual Event, USA.

© 2021 Copyright held by the owner/author(s). Publication rights licensed to ACM.
ACM ISBN 978-1-4503-8479-7/21/10...\$15.00
<https://doi.org/10.1145/3462204.3481769>

1 INTRODUCTION

The COVID-19 coronavirus outbreak has spread worldwide, disrupting all aspects of our lives. Mobile technology and mHealth solutions has been leveraged in numerous communities to control the spread of the coronavirus [2, 12]. Social pandemic-tracking apps support several functionalities such as contact tracing, self-assessments of symptoms, and enforcing social distancing, which are used to help prevent the spread of the coronavirus [9]. The use of such applications in several communities were mandatory to engage in any social activity [17]. Older adults within such communities where overwhelmed with the amount of restrictions and technologies they need to learn and use in a short period of time to engage in any basic daily social activity, such as going to grocery shops, hospitals, or attending religious gatherings, negatively impacting their physical and mental well-being [6, 12].

Several studies have been conducted amid the COVID-19 outbreak focusing on technology use and mHealth [2, 9, 12], learning [7, 23], and social issues and privacy concerns [3, 7, 10, 19, 24, 31]. We see a need to explore and deeply understand the challenges faced by older users of such social applications. To explore these challenges and issues during the COVID-19 pandemic, we conducted an online survey in early 2021 and examined the adoption of social pandemic tracking applications among older adults and their families and the challenges they face. We also performed one-on-one online interviews with older adults and undertook several unstructured observations of their experiences.

Our initial findings show that most older users of such mandatory social pandemic tracking applications required help from others to use these applications, increasing the dependence on others and the chance of contracting the coronavirus from cross-contamination caused by close contact and device sharing with family members and strangers. Such initial results make us question the level of technology literacy we reached within our communities, and call for new and innovative ways to help older adults easily adopt such technology. Our findings provide important implications for understanding current challenges in using such applications, providing initial insights on how to develop better future social pandemic tracking applications and improving the experiences and well-being of older adults.

2 BACKGROUND

Mobile technology adoption has become more apparent with the COVID-19 coronavirus outbreak [2]. Due to the social isolation

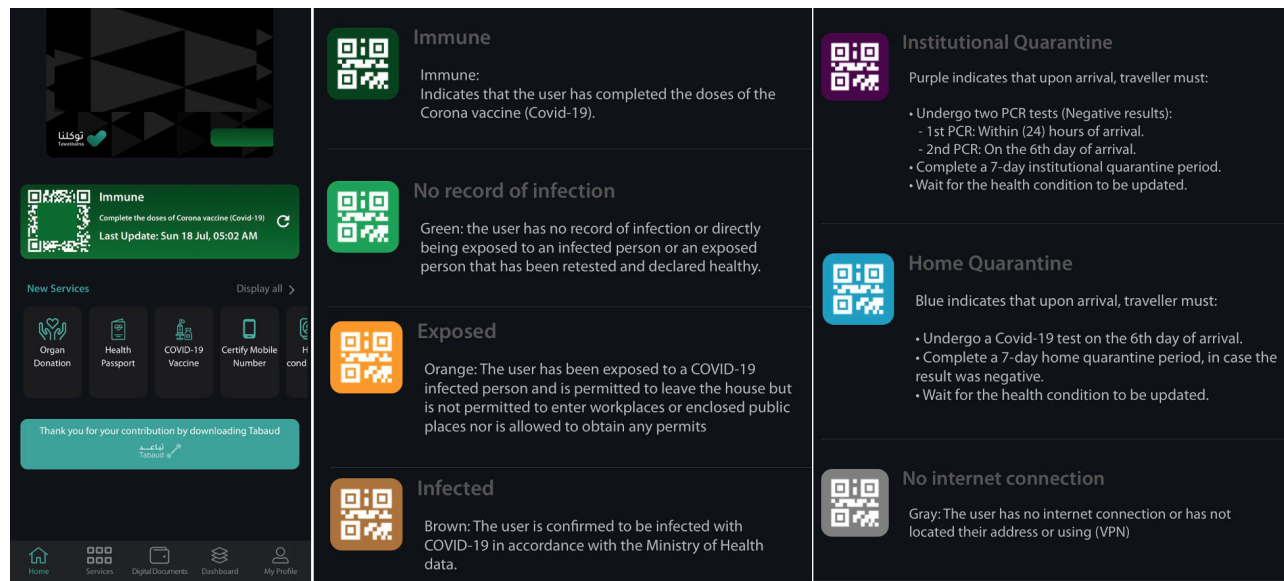


Figure 1: Social Pandemic-Tracking App (Tawakkalna) used in Saudi Arabia for contact tracing and exposure control. It is a mandatory application that must be used by every adult in order to enter or engage in any social activity (e.g., going to a grocery shop, universities, etc.) to help prevent the spread of the COVID-19 coronavirus. (Screenshots taken by Sultan A. Alharthi).

imposed on our communities to help prevent the spread of the coronavirus, businesses, education systems, and health care providers urgently adopted remote and mobile technology to overcome this isolation [12]. Such urgent and unplanned adoption caused a disruption in how we live our lives and how we engage in any social activity, especially impacting older adults, low income families, and other minorities [14, 25].

Technology & Health Literacy. Results of prior research on the effectiveness of mHealth applications used by older adults show promise [5, 22, 29]. However, these previous studies are mostly about mHealth use and experiences under normal circumstances. Pandemic-driven technology requires users to have immediate access to technology, including internet connections and smart devices, along with the knowledge needed to use such technology. This fast-paced shift toward digitalization exacerbated the digital divide in less prepared communities and minority, exposing digital inequities globally [21, 32]. The intersection of technology and health literacy has been discussed previously [8, 26, 30]. Prior research show that low technology literacy is a barrier to mHealth adoption [27]. Along with low technology literacy, the lack of understanding new and unfamiliar health crisis and its consequences exacerbate the situation and hinder mitigation efforts [26]. Urgent technology and health literacy interventions are currently needed.

Pandemic-Driven Technology. Research investigating pandemic-driven technology is growing [2, 12]. New and innovative COVID-19 mobile apps and solutions has been introduced within the past 15 months to help prevent the spread of the coronavirus (e.g., Tawakkalna [9] (Figure 1), Health Code [33], and NHS COVID-19 [18]). However, the adoption and use of such technology comes with its own challenges. Social issues and privacy concerns are

being discussed within the HCI and CSCW communities with respect to such pandemic-driven technology [3, 7, 10, 19, 24, 31]. Hidayat-ur-Rehman et al. [9] evaluated the performance of three COVID-19 mobile apps used in Saudi Arabia. The findings show that such applications are effective and their performance is good. However, the authors suggest that “More awareness campaigns are needed so that more people adopt the use of these applications, which could significantly help in identification of new cases” [9]. Designers, decision-makers, and researchers need to take into consideration all these concerns and recommendations when designing and imposing such technology.

Older Adults and the Pandemic. In Saudi Arabia, older adults are individuals who are 65 years of age or older [20]. This age group is becoming increasingly vulnerable due to the pandemic [6]. Loneliness and mental health issues of older adults increased during the pandemic due to enforced social isolation and quarantine [28]. Prior research show that understanding older users experiences can help in designing better technology and increase the adoption of it [11–13, 16]. Kondylakis et al. [12] conducted a thorough systematic review of COVID-19 mobile apps literature. The authors argue that: “Understanding the needs of older individuals in the COVID-19 pandemic period would be the first necessary step toward designing and developing mobile apps to encourage their physical and mental well-being” [12]. In this work, we see a need to further explore and deeply understand the challenges faced by older user of such social applications, especially with the increased integration of such technology into all areas of our lives. Insights from this work can help support designers and health care providers in understanding the needs of this vulnerable population.



Figure 2: A number of young and older adults waiting in line to attend a social event, in which health inspectors at the entrance check each person's exposure status indicated on the mandatory pandemic-tracking application (Tawakkalna) used in Saudi Arabia (see Figure 1). The app shows whether a user is infected or not. Health inspectors at the entrance also check mask-wearing, take body temperature, and enforce social distancing. Any person without this application would be denied entry to such social events.

3 METHODOLOGY

In this work, we examined the adoption of mandatory social pandemic tracking mobile applications among older adults and their families and the challenges they face in Saudi Arabia. We first developed our own survey by reviewing prior studies that take into account working with older adults and their families [16], privacy and social issues within the MENA region [1], and similar COVID-19 studies [9, 24]. Participants were recruited through different social media platforms (i.e., Twitter, Facebook, and WhatsApp). The survey collected information about the use of such technology and the challenges older adults face. The survey included questions about the following: name of applications used, type of application being used, help and support needs, barriers, and challenges. After we collected the survey data, we conducted several one-on-one online interviews with older adults to further understand the challenges they face. Finally, we conducted multiple unstructured observations of how older adults use such technology in order to understand their experiences taking into account physical and socio-cultural factors (Figure 2). Such unstructured observations helped uncover some issues and challenges that were not obvious to our participants in the surveys and interviews. These three data sources complement each other and provided a multifaceted view of the issues and challenges older adults and their families face in using social pandemic tracking mobile applications. After all data were collected from these three sources, we analyzed the data to identify the main challenges faced by older adults and their families. We followed a thematic analysis approach [4], looking for similar themes and patterns among our data.

A total of 397 users participated in the survey and 5 older adults participated in the online interviews. Older adults in this study

ranged in age between 60 and 92 years old ($M = 65$, $SD = 12.98$). The survey focused on older adults' families to gain some understanding from them in regards to how older adults living with them experience such applications. Families also helped us reach and conduct the one-on-one interviews with the older adults since conducting research with this age group is difficult, especially during the pandemic due to social distancing and health concerns. Based on the survey, 80% of all participants have older adults living with them. This is a common living situation in most MENA regions, in which older adults live with their families in the same household. 69% of these older adults use a smartphone on regular basis.

4 FINDINGS AND DISCUSSION

Our initial findings show that older adults faced some challenges in using social pandemic tracking applications. Due to these challenges, they usually end up asking for help from others to use these applications, increasing the dependence on others and the chance of contracting the coronavirus.

Application Use and Help Needs. When participants were asked about the applications they usually need help with, they mentioned a range of different pandemic related application, 64% mentioned Tawakalna, 30% mentioned Sehaty, and 16% mentioned Tabaoud. All of these applications are used within Saudi Arabia for pandemic related services, such as contact tracing, entering any social gatherings, and vaccination appointments and are mandatory to be used by every adult. Other applications also were mentioned that required help, such as WhatsApp 47% and Absher 37% (Absher is an e-government application used in Saudi Arabia providing a range of government related services). When families were asked about help requests, 82% of participant mentioned that such pandemic related

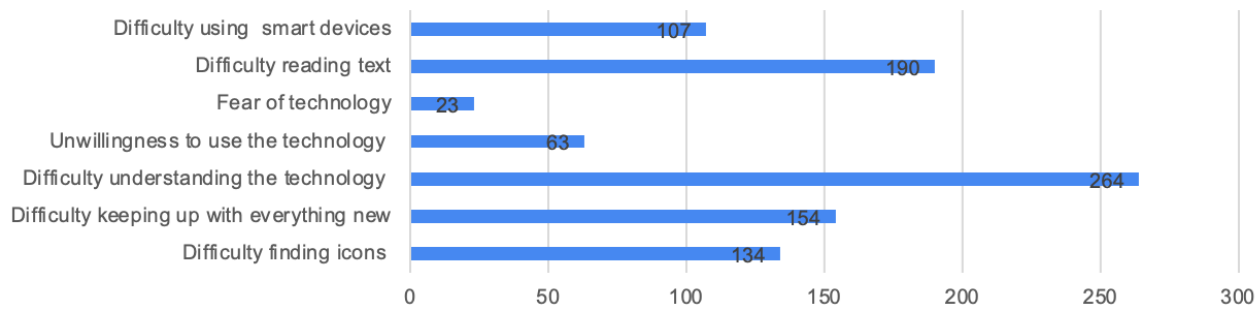


Figure 3: Results of the main challenges older adults face using social pandemic tracking applications.

application are difficult to use by older adults living with them. 83% of participants mentioned that older adults have previously asked them multiple times for help in order to use such applications.

Challenges. When families were asked about the challenges that hinder older adults from using such applications by themselves, they mentioned a range of issues (Figure 3), including: difficulty understanding the technology 66% (264/397), difficulty reading text 48% (190/397), difficulty keeping up with everything new 39% (154/397), difficulty finding icons 34% (134/397), difficulty using smart devices 27% (107/397), unwillingness to use such technology 16% (63/397), and fear of the technology 6% (23/397).

We observed that such difficulties impact older adults from using these application on their own when trying to engage in physical social activities, in some cases forcing them to ask strangers to assist them resulting in an increased dependence on others. These difficulties result in excluding them from engaging in any basic physical social activities since the use of these applications are mandatory, thus exacerbating social inequities. During the interviews, one participant mentioned that “*I need someone to use it for me because I don’t know how*” [P1], others also mentioned that “*I write all my passwords in a paper, or let my daughters memorize it for me*” [P2]. While the intended benefit of this type of applications is to organize physical social activities and maintain social distancing to reduce the spread of the coronavirus, older adults and less technology-literate people are the ones who are mostly disadvantaged due to these challenges. When entering any physical social activities, we observed in several cases older adults and domestic workers requesting help from others to assist them with such applications. Due to such challenges, they often end up gathering at the entrance where the services are provided trying to figure out how to use the application or fix any technical issues, increasing their chances of contracting the virus. Unfortunately, all these challenges reported and observed are affecting mostly older adults, who are most vulnerable to COVID-19, thus defeating the main purposes of these pandemic-driven technologies and solutions. New and innovative approaches need to be investigated to design better pandemic related technologies that take into account the challenges older adults face.

5 CONCLUSION AND FUTURE WORK

Our results indicate that older adults face a number of challenges in using pandemic tracking applications. These results show an urgent need for further work on improving the design and usability of

such technologies used by older adults. HCI and CSCW researchers and designers could use such initial insights to improve current pandemic-driven solutions, especially with the increased integration of digital technology into all areas of our lives. For future work, engaging older adults and less technology-literate people in the design process of such pandemic-driven technologies and solutions is a necessary step in designing better usable solutions [12, 15]. We also see a need to perform several usability studies on different pandemic tracking applications to evaluate them and provide suggestions on how to improve them.

REFERENCES

- [1] Norah Abokhodair and Sarah Vieweg. 2016. Privacy & Social Media in the Context of the Arab Gulf. In *Proceedings of the 2016 ACM Conference on Designing Interactive Systems* (Brisbane, QLD, Australia) (DIS ’16). Association for Computing Machinery, New York, NY, USA, 672–683. <https://doi.org/10.1145/2901790.2901873>
- [2] Emmanuel Abidemi Adeniyi, Joseph Bamidele Awotunde, Roseline Oluwaseun Ogundokun, Paul Oluwatoba Kolawole, Moses Kazeem Abiodun, and Abosede Adepeju Adeniyi. 2020. Mobile health application and COVID-19: Opportunities and challenges. *Journal of Critical Reviews* 7, 15 (2020), 3481–3488. <http://www.jcreview.com/fulltext/197-1599652532.pdf>
- [3] Ala Sarah Alaqra and Agnieszka Kitkowska. 2021. Impact of Intrinsic Factors and COVID-19 Pandemic on the Use of Technology. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA ’21). Association for Computing Machinery, New York, NY, USA, Article 466, 7 pages. <https://doi.org/10.1145/3411763.3451669>
- [4] Virginia Braun and Victoria Clarke. 2012. Thematic analysis. In *APA handbook of research methods in psychology, Vol 2: Research designs: Quantitative, qualitative, neuropsychological, and biological*. American Psychological Association, Washington, DC, US, 57–71. <https://doi.org/10.1037/13620-004>
- [5] Maryam Changizi and Mohammad H Kaveh. 2017. Effectiveness of the mHealth technology in improvement of healthy behaviors in an elderly population—a systematic review. *Mhealth* 3 (2017), 1–9. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5803024/>
- [6] Centers for Disease Control and Prevention. 2020. Older unvaccinated adults are more likely to be hospitalized or die from COVID-19. <https://www.cdc.gov/coronavirus/2019-ncov/need-extra-precautions/older-adults.html>
- [7] ADAM GARCIA, Garrett B Powell, Davis Arnold, Luis Ibarra, Matthew Pietrucha, Michael Kelland Thorson, Abigail Verhelle, Nathan B Wade, and Samantha Webb. 2021. Learned Helplessness and Mental Health Issues Related to Distance Learning Due to COVID-19. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA ’21). Association for Computing Machinery, New York, NY, USA, Article 480, 6 pages. <https://doi.org/10.1145/3411763.3451526>
- [8] Muger V. Geana, Sherri Anderson, Ashlyn Lipnick, Joi L. Wickliffe, and Megha Ramaswamy. 2021. Managing Technology, Content, and User Experience: An mHealth Intervention to Improve Women’s Health Literacy after Incarceration. *Journal of Health Care for the Poor and Underserved* 32, 2 (2021), 106–127. <https://doi.org/10.1353/hpu.2021.0053> Publisher: Johns Hopkins University Press.
- [9] Imdadullah Hidayat-ur Rehman, Arshad Ahmad, Manzoor Ahmed, and Aftab Alam. 2021. Mobile Applications to Fight against COVID-19 Pandemic: The Case of Saudi Arabia. *TEM Journal* 10 (Feb. 2021), 69–77. <https://doi.org/10.18421/TEM101-09>

- [10] Megan Hofmann, Udaya Lakshmi, Kelly Mack, Scott Hudson, Rosa Arriaga, and Jennifer Mankoff. 2021. The Right to Help and the Right Help: Fostering and Regulating Collective Action in a Medical Making Reaction to COVID-19. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. Association for Computing Machinery, New York, NY, USA, 1–13. <https://doi.org/10.1145/3411764.3445707>
- [11] Nancy J. Karlin, Joyce Weil, and Wejdan Felmban. 2016. Aging in Saudi Arabia: An Exploratory Study of Contemporary Older Persons Views About Daily Life, Health, and the Experience of Aging. *Gerontology and Geriatric Medicine* 2, 0 (2016), 1–9. <https://doi.org/10.1177/2333721415623911>
- [12] Haridimos Kondylakis, Dimitrios G Katehakis, Angelina Kouroubali, Fokion Logothetidis, Andreas Triantafyllidis, Ilias Kalamaras, Konstantinos Votis, and Dimitrios Tzovaras. 2020. COVID-19 Mobile Apps: A Systematic Review of the Literature. *J Med Internet Res* 22, 12 (9 Dec 2020), e23170. <https://doi.org/10.2196/23170>
- [13] Roland Kunz and Markus Minder. 2020. COVID-19 pandemic: palliative care for elderly and frail patients at home and in residential and nursing homes. *Swiss Medical Weekly* 150 (March 2020), w20235. <https://doi.org/10.4414/smww.2020.20235>
- [14] Ingrid D. Lui, Nimisha Vandan, Sara E. Davies, Sophie Harman, Rosemary Morgan, Julia Smith, Clare Wenham, and Karen Ann Grepin. 2021. We also deserve help during the pandemic: The effect of the COVID-19 pandemic on foreign domestic workers in Hong Kong. *Journal of Migration and Health* 3 (2021), 100037. <https://doi.org/10.1016/j.jmh.2021.100037>
- [15] Hanuma Teja Maddali, Emma Dixon, Alisha Pradhan, and Amanda Lazar. 2020. Supporting Remote Participation When Designing with People with Dementia. In *Conference Companion Publication of the 2020 on Computer Supported Cooperative Work and Social Computing* (Virtual Event, USA) (CSCW '20 Companion). Association for Computing Machinery, New York, NY, USA, 335–340. <https://doi.org/10.1145/3406865.3418316>
- [16] Soud Nassir and Tuck Wah Leong. 2017. Traversing Boundaries: Understanding the Experiences of Ageing Saudis. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems* (Denver, Colorado, USA) (CHI '17). Association for Computing Machinery, New York, NY, USA, 6386–6397. <https://doi.org/10.1145/3025453.3025618>
- [17] Arab News. 2021. Mandatory use of Tawakkalna app to access public places gets mixed reaction in Saudi Arabia. <https://arab.news/2vy3s>
- [18] NHS. 2021. The NHS COVID-19 app support website. <https://www.covid19.nhs.uk/>
- [19] Shuo Niu, Ava Bartolome, Cat Mai, and Nguyen Binh Ha. 2021. #StayHome #WithMe: How Do YouTubers Help with COVID-19 Loneliness?. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 338, 15 pages. <https://doi.org/10.1145/3411764.3445397>
- [20] General Authority of Stat. 2019. Results of the Elderly Survey 2019. https://www.stats.gov.sa/sites/default/files/repport_compressed_1_0.pdf
- [21] United Nations Conference on Trade and Development. 2021. Coronavirus reveals need to bridge the digital divide | UNCTAD. <https://unctad.org/news/coronavirus-reveals-need-bridge-digital-divide>
- [22] Merja Sallinen, Outi Henttonen, and Anne Karki. 2015. Technology and active agency of older adults living in service house environment. *Disability and Rehabilitation: Assistive Technology* 10, 1 (2015), 27–31. <https://doi.org/10.3109/17483107.2013.836685>
- [23] Samantha L. Schneider and Martha Laurin Council. 2021. Distance learning in the era of COVID-19. *Archives of Dermatological Research* 313, 5 (July 2021), 389–390. <https://doi.org/10.1007/s00403-020-02088-9>
- [24] John S. Seberger and Sameer Patil. 2021. Us and Them (and It): Social Orientation, Privacy Concerns, and Expected Use of Pandemic-Tracking Apps in the United States. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 65, 19 pages. <https://doi.org/10.1145/3411764.3445485>
- [25] Shreela V Sharma, Ru-Jye Chuang, Melinda Rushing, Brittni Naylor, Nalini Ranjit, Mike Pomeroy, and Christine Markham. 2020. Social Determinants of Health-Related Needs During COVID-19 Among Low-Income Households With Children. *Preventing chronic disease* 17 (2020), 119–126. <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7553207/>
- [26] Benjamin Smith and Jared W. Magnani. 2019. New technologies, new disparities: The intersection of electronic health and digital health literacy. *International Journal of Cardiology* 292 (2019), 280–282. <https://doi.org/10.1016/j.ijcard.2019.05.066>
- [27] Austin J Triana, Roman E Gusdorf, Kaustav P Shah, and Sara N Horst. 2020. Technology literacy as a barrier to telehealth during COVID-19. *Telemedicine and e-Health* 26, 9 (2020), 1118–1119. <https://www.liebertpub.com/doi/pdfplus/10.1089/tmj.2020.0155>
- [28] Theo G van Tilburg, Stephanie Steinmetz, Elske Stolte, Henriette van der Roest, and Daniel H de Vries. 2020. Loneliness and Mental Health During the COVID-19 Pandemic: A Study Among Dutch Older Adults. *The Journals of Gerontology: Series B* 7 (08 2020), 249–255. <https://doi.org/10.1093/geronb/gbaa111>
- [29] Kerryellen G. Vroman, Sajay Arthanat, and Catherine Lysack. 2015. Who over 65 is online? Older adults dispositions toward information communication technology. *Computers in Human Behavior* 43 (2015), 156–166. <https://doi.org/10.1016/j.chb.2014.10.018>
- [30] Ronie Walters, Stephen J. Leslie, Rob Polson, Tara Cusack, and Trish Gorely. 2020. Establishing the efficacy of interventions to improve health literacy and health behaviours: a systematic review. *BMC Public Health* 20, 1 (June 2020), 1040. <https://doi.org/10.1186/s12889-020-08991-0>
- [31] Yun Wang, Ying Liu, Weiwei Cui, John Tang, Haidong Zhang, Doug Walston, and Dongmei Zhang. 2021. Returning to the Office During the COVID-19 Pandemic Recovery: Early Indicators from China. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI EA '21). Association for Computing Machinery, New York, NY, USA, Article 417, 6 pages. <https://doi.org/10.1145/3411763.3451685>
- [32] Tamika Williams, Robert McIntosh, and William Russell. 2021. Equity in Distance Education During COVID-19. *Research in Social Sciences and Technology* 6, 1 (May 2021), 1–24. <https://doi.org/10.46303/ressat.2021.1>
- [33] Sophia L. Zhou, Xianhan Jia, Samuel Skinner, William Yang, and Isabelle Claude. 2021. Lessons on Mobile Apps for COVID-19 from China. *Journal of Safety Science and Resilience* 2 (2021), 40–49. <https://doi.org/10.1016/j.jnlssr.2021.04.002>